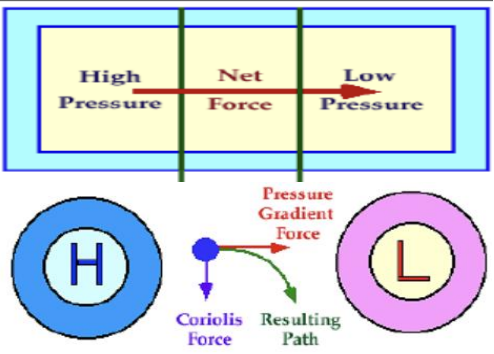
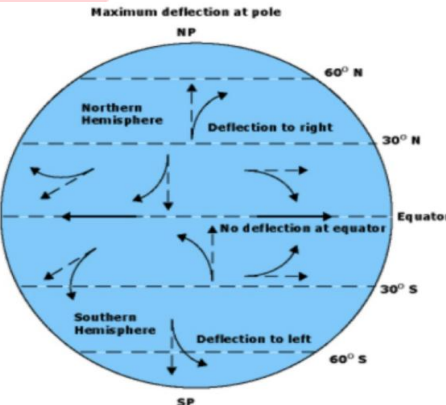


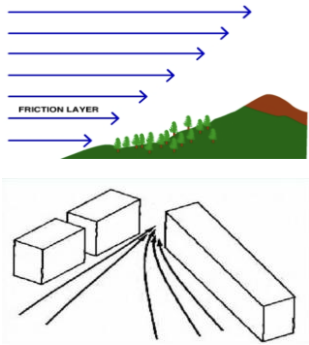
GALLANT IAS ACADEMY
PRELIMS CUM MAINS BATCH 2027
GEOGRAPHY
CLIMATOLOGY
MODULE 4
PRESSURE SYSTEMS

Wind Systems

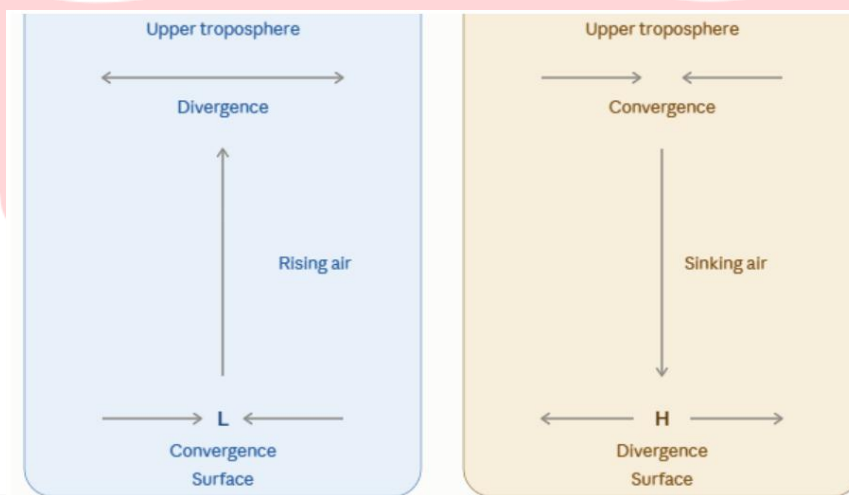
The horizontal movement of air is called **wind**. The vertical movement of air is called **current**. There are ascending (updraft) and descending (downdraft) air currents. The air near the Earth's surface moves from **high-pressure to low-pressure** regions. Winds represent nature's attempt to balance out the uneven distribution of air pressure over the Earth's surface. As a convention, wind is always named as per the direction of its origin.

FORCES AFFECTING WIND SYSTEMS

Force	Definition & Key Points
1. Pressure Gradient Force	Created due to differences in atmospheric pressure. The difference of pressure between any two places is called the pressure gradient (also called barometric slope). It is the basic activating force of wind - it initiates the movement of air particles. Acts at right angles to the isobar, in the direction of low pressure. [An isobar is a line connecting places having the same pressure.] Stronger where isobars are placed closer together. 
2. Coriolis Force	All moving objects on Earth appear to be deflected because of the Earth's rotation. Ferrel's law of apparent deflection: deflection to the right in the northern hemisphere and to the left in the southern hemisphere. Increases with the speed of the moving body (increases with wind velocity). Negligible (zero) near the equator; increases towards the poles (higher latitudes). 
3. Frictional Force	Force generated by the resistance of a surface against a moving object. Acts to slow down wind movement and modify its direction. Minimum over sea surfaces. Higher over land surfaces with roughness (vegetation, hills, buildings). The zone of the lower atmosphere where frictional force becomes effective is called the friction layer - maximum at the surface, decreasing upward. The funnelling effect: wind forced

	through a narrow area increases in speed (explains record rainfall at Cherrapunji and Mawsynram). 
4. Centrifugal Force	Acts on winds moving along a curved path. Directed outward from the centre of the curve. A factor in high-velocity winds operating in violent tropical cyclones.

AIR PRESSURE



The atmosphere is a mixture of several gases and has its own weight. This weight exerted by the atmosphere is responsible for atmospheric pressure. (Technically, air pressure is defined as the total weight of a column of air above a unit area at sea level.) High-pressure systems are often created by sinking air packets - also called **highs**. Low-pressure systems are often created by rising air packets - also called **lows**.

A **zone of convergence** is a region with a net inflow of air. Low-pressure zones make zones of convergence. A **zone of divergence** is a region with a net outflow of air. High-pressure zones make zones of divergence. For a zone of convergence (L) near the Earth's surface, there is a zone of divergence (H) in the upper troposphere, and vice versa.

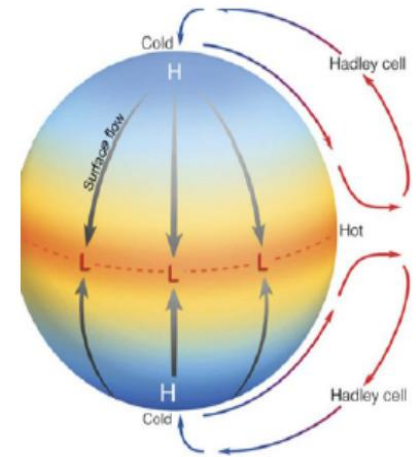
ATMOSPHERIC CIRCULATION MODELS

1. Unicellular Model (Hadley's Model / Thermal Model)

By **George Hadley (18th century)**. Based on three assumptions: (i) the sun is overhead at the equator; (ii) the Earth's surface is uniform (either all land or all water); and (iii) the Earth is non-rotating (Coriolis force is excluded). At the equator, warm air rises, creating the equatorial low pressure at the surface. The rising air diverges in the upper troposphere and moves towards the poles. Over the poles it cools and sinks, creating the polar high pressure. At the poles, the cold air diverges from the polar high pressure and moves towards the equator, where it warms and rises again.

Thermal model: According to Hadley, intense heating at the equatorial region creates a low pressure, and intense cooling at the polar region results in a high pressure. The driving force is the thermal gradient across the latitudes. It is also called the thermal model.

Unicellular model: Based on a single, thermally induced convective cell operating between the equator and the poles - one circulating cell per hemisphere.



2. Tricellular Model (Ferrel's Model / Dynamic Model)

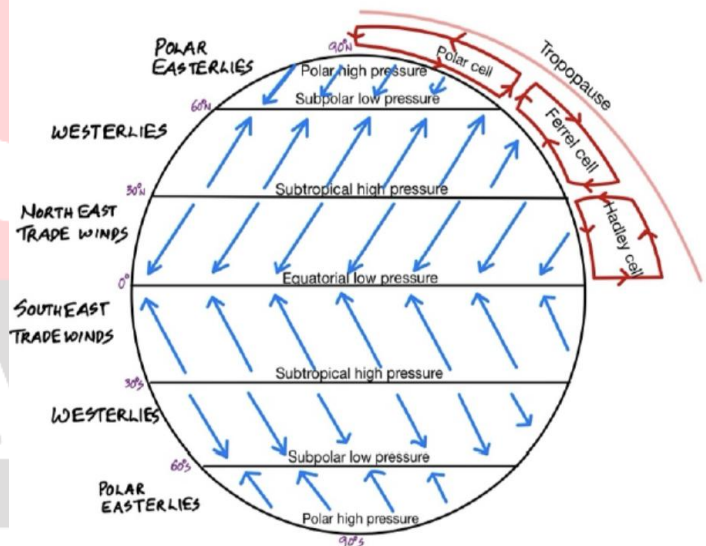
By **William Ferrel (19th century)**. Based on three considerations: (i) the sun is overhead at the equator; (ii) the Earth's surface is uniform; and (iii) the Earth is rotating (the Coriolis force is taken into consideration).

Hadley cell (Tropical cell): Warm air rising from the equator diverges from the high pressure in the upper troposphere, and while moving towards the poles the wind is deflected by the Coriolis force. By the time the winds reach 30° latitude (the subtropics) they cool and sink, creating the subtropical high pressure. From the sinking air at the subtropical high pressure, part of the wind system diverges towards the equator (trade winds), completing the Hadley cell. Part of the diverging air also moves towards higher latitudes (westerlies).

Ferrel cell (Midlatitude cell or Polar front cell) and Polar cell (Subpolar cell): At the poles, sinking cold air creates the polar high pressure. The diverging

wind systems from the polar high pressure (polar easterlies) meet the air coming from the subtropics (westerlies) around 60° latitude (the subpolar latitudes). After the meeting of air packets from the polar and subtropical regions, the air is forced to rise, creating the subpolar low pressure. The rising air from the subpolar region diverges at the tropopause, completing the Polar cell and the Ferrel cell.

Tricellular model: Since there are three circulating cells in each hemisphere, Ferrel's model is also called the tricellular model. The Hadley and Polar cells are driven directly by different effects of surface heating - they are **thermally direct cells**. The Ferrel cell is called a **thermally indirect cell**.



PRESSURE PATTERNS ON THE EARTH

Vertical Pressure Pattern

With increasing height in the atmosphere, density decreases. This results in a corresponding decrease in pressure with increasing elevation.

Horizontal Pressure Pattern Accounting for the Land-Water Differential

Planetary circulations and pressure belts are much more complex than in Ferrel's model. Ferrel's continuous zonal pressure belts are possible only on a surface that is homogeneous. Since the Earth's surface is not uniform, the pressure belts are not continuous straight - it bends.

The **polar high pressure** is a continuous belt in both the south and north because of the presence of ice cover (a homogeneous surface). The **subpolar low pressure** of the southern hemisphere is a continuous pressure belt because of the dominance of the ocean surface. The **equatorial low pressure** is also a continuous belt, since water dominates the equatorial region and the equatorial region practically has no seasons. The disruption of pressure belts is **greater in the northern hemisphere** because the north has a more complex mix of land and water.

Shifting of the Pressure Belts

Since the sun is not stationary over the equator and has an apparent movement between the equator and the tropics, the pressure belts (or cells) shift along with the movement of the sun. In **June**, when the overhead sun is over the Tropic of Cancer, all the belts move about **5°-10°N** of their average position. In **December**, when the sun is overhead at the Tropic of Capricorn, all pressure belts shift **5°-10°S**. The shift of the pressure belts is greatest in the lower latitudes and least in the polar regions.

MAJOR PRESSURE BELTS OF THE WORLD

Pressure Belt	Location	Type	Key Characteristics
Equatorial Low / ITCZ / Doldrums	5°N to 5°S	Thermally induced	Zone of convergence of trade winds from both hemispheres. Hot, rising air. Convictional rainfall, cumulonimbus clouds. Belt of Calm - near absence of surface winds. Also called the thermal equator.
Subtropical High / Horse Latitudes	25°-35° (around 30°)	Dynamically induced	Due to subsidence of air from the equatorial region. Warm and dry - most of the world's hot deserts lie here. Calm, descending air currents / wind divergence. Horse latitudes: horses thrown overboard when ships were becalmed.
Subpolar Low	60°-65° latitude	Dynamically induced	Polar easterlies meet westerlies around 60°; air is forced to rise, creating

			low pressure. Temperatures are low year-round - a high pressure would be thermally expected, but dynamic forces create a low.
Polar High	80°-90°N and S	Mostly thermally induced	Cold, heavy air subsides on reaching the poles. Small in area, extending around the poles.

PLANETARY WINDS

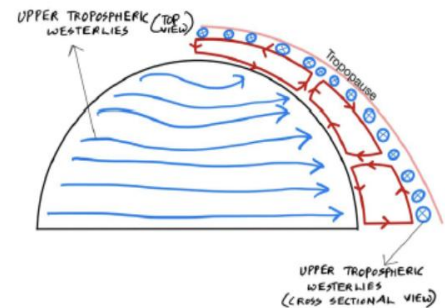
Trade winds, westerlies, and polar easterlies together constitute **planetary winds**. They are global in extent, hence called planetary winds. They are also called **permanent winds** because they maintain the same direction throughout the year (also known as prevailing winds, invariable winds, and primary circulations). They are generated by pressure gradient force and affected by the Coriolis force, and are extremely important as they carry water vapour from the oceans to the continents and transfer heat energy from tropical areas towards the poles.

Wind	Direction	Key Features	Effects
Trade Winds	Subtropical high (30°) to equatorial low (ITCZ)	Most regular of all planetary winds - highly consistent in direction and velocity. Named for helping sea merchants sail their ships.	Bring heavy rainfall to tropical east coasts. Offshore on west coasts - hot deserts form (Atacama, Mojave, Sahara, Namib, Great Australian).
Westerlies	Subtropical high (30°) to subpolar low (60°)	SW in NH, NW in SH. Highly variable in velocity and direction (temperate cyclones embedded within them).	Onshore on west coasts (40°-60°). In SH: Roaring Forties, Furious Fifties, Shrieking Sixties.
Polar Easterlies	Polar high to subpolar low	High velocity, cold, dry. Variable. More regular in SH due to continuous pressure belts.	Carry snow particles, reduce visibility, create cold-wave conditions. Can reach lower latitudes during a polar outbreak.

UPPER TROPOSPHERIC WINDS

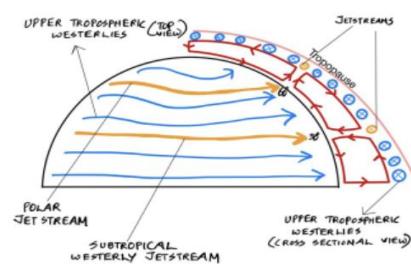
Upper Tropospheric Westerlies

The upper tropospheric winds (aloft winds) are not part of the vertical meridional cells suggested by Ferrel's model. Conditions in the upper troposphere are not exactly the mirror opposite of those at the surface, so aloft winds are independent of surface winds. In the upper troposphere, a **high pressure** develops over the equator and a **low pressure** over the poles, so a single pressure gradient force drives the winds from the equator to the poles. Because the atmosphere is rarefied there, winds experience little friction and attain great speeds. The Coriolis force therefore becomes very strong and deflects the winds until they move parallel to the isobars - this is called the **geostrophic condition**, and such winds are called **geostrophic winds**. Therefore, winds in the upper troposphere are found to be **westerly at all latitudes**.



Jet Streams

Jet streams are **narrow bands of swift-moving winds in the upper atmosphere**. The core of a jet stream has wind speeds exceeding **300 km/hr**. They are embedded in the upper tropospheric westerlies at the junctions of the Hadley, Ferrel, and Polar cells. Jet streams follow a **meandering path** to conserve angular momentum. They are characterised by **seasonal variability**, with highest velocity during winters. Jet streams help explain many climatic phenomena such as winter rainfall in India, the south-west monsoon, and the initiation of temperate cyclones.



Types of Jet Streams

Jet Stream	Location	Nature	Key Features
Polar Jet Stream (Polar Front Jet Stream)	55°-65° latitude; junction of Polar and Ferrel cells	Permanent	Formed above subpolar latitudes above the convergence zone of polar cold air mass and tropical warm air mass. Embedded within Rossby waves and oscillates through its index cycles.
Subtropical Westerly Jet Stream	30°-35° latitude; junction of Hadley and Ferrel cells	Permanent	Moves in the upper troposphere above the subtropical high-pressure belt. Circulates from west to east more regularly than the polar front jet streams.

Polar Night Jet Stream / Stratospheric Sub-Polar Jet Stream	Around the poles, in the stratosphere	Temporary (seasonal)	Develops in the winter season. Very strong westerly circulation with high wind velocity during winters; velocity decreases during summers.
Tropical Easterly Jet Stream	Over India and Africa during summer; along Allahabad-Mangalore axis	Temporary (seasonal/regional)	Develops due to intense heating of the Tibetan Plateau, creating Tibetan low pressure and upper-atmospheric divergence. Sinks near the Mascarene high pressure near Madagascar. Mostly affecting South Asia.

LOCAL WINDS

Local winds are produced by local variability in temperature and pressure conditions, so they are confined to certain localities. Despite their limited extent, their impact is highly pronounced in the areas where they flow.

1. Periodical Local Winds (Direction reverses within 24 hours)

Wind	Direction / Timing	Key Features
Sea Breeze	Sea to land - during the day	Land heats up faster than water, creating local low pressure over land. Cool, moist air flows in from the sea. Onshore wind.
Land Breeze	Land to sea - at night or early morning	Land loses heat faster and becomes cooler than the sea. Pressure gradient runs from land to sea. Offshore wind.
Valley Breeze (Anabatic Wind)	Valley to mountain slopes - during the day	The sun heats mountain slopes faster than shaded valleys. Warm air rises; air from the valley blows up the slopes. Reaches mountain peaks, yields precipitation through cumulus clouds.
Mountain Breeze (Katabatic Wind)	Mountain slopes to valley - at night	Slopes cool through radiation; cool air descends the valley slopes. Causes temperature inversion in valleys - valley floors see frost while upper slopes remain frost-free.

2. Non-Periodical Local Winds - Warm (Hot)



Wind	Region	Key Features
Chinook	Rocky Mountains, USA and Canada	Warm, moist air rises windward slope, cools and gives orographic rainfall; crosses mountain, descends leeward slope compressed and heated. Called 'the snow eater' - rapid temperature rise melts snow, clearing fields for winter crops.
Santa Ana	Southern California, USA	Warm, dry, strong, dusty wind from the north-east or east over southern California. A channelled chinook-type wind funnelled into California's Santa Ana Valley. Dries out vegetation, raising wildfire risk.
Black Roller	Great Plains, USA	Regional name for a warm, dry, dust-laden wind of the harmattan type.
Zonda	Andes, Argentina	Chinook-like warm wind (same foehn-type mechanism) on the eastern slope of the Andes.
Harmattan	Sahara Desert, West Africa	Warm, dry, dusty winds blowing from the north-east over the Sahara. Extremely dry, pick up sand - can cause dust storms. Called the 'doctor' along the Guinea coast - brings dry, pleasant weather after warm, moist conditions. Actually a special type of north-east trade wind.
Sirocco	Sahara Desert to Mediterranean and Europe	Warm, dry, dusty wind blowing north from the Sahara. Also called 'blood rain' for reddish Saharan sand it carries. Local names: khamsin (Egypt), gibli (Libya), chili (Tunisia); leveche and leste in Spain and Canary/Madeira Islands.

Loo	Northern plains of India and Pakistan	Hot, dry wind in April, May and June, especially in afternoons. Originates in the Thar Desert, sweeps from north-west and west towards east. Temperature ranges from 45°C to 50°C. Notorious in Rajasthan, Haryana, Punjab, UP and Delhi.
Shamal	Mesopotamia and the Persian Gulf	Regional name for a warm, dry, dust-laden wind of the harmattan type.
Foehn	Alps, Switzerland	Same foehn-type mechanism as chinook. Temperatures range from 15°C to 20°C. Melts snow in valleys; Switzerland's valleys are called a 'climatic oasis' in winter. Helps grapes ripen early.
Brickfielder	Victoria, Australia	Regional name for a warm, dry, dust-laden wind of the harmattan type.
Nor'wester	New Zealand (eastern side of Southern Alps)	Regional name for a warm, dry, dust-laden wind of the harmattan type.

3. Non-Periodical Local Winds - Cold

Wind	Region	Key Features
Papagayo	Pacific coast of Central America (Guatemala to Costa Rica)	Strong, north-easterly wind. Produced by a cold air mass travelling down through the Central American mountains.
Blizzard	Polar regions, Siberia, Canada, USA	Violent, stormy, cold, powdery polar wind laden with snow. Causes sudden drop in temperature to sub-freezing levels, thick snow cover, and cold waves. Velocity can reach close to 100 km/hr. Called 'Norther' in southern USA; 'Burran' in Siberia.
Pampero	Argentina and Uruguay	Severe squalls in Argentina and Uruguay, particularly the Rio de la Plata area. Usually forms in winter, associated with marked cold fronts, accompanied by sharp temperature drop and sudden change in wind direction.
Burran	Siberia	Regional name for the same violent, cold, snow-laden polar wind described under Blizzard.
Mistral	Alps, through Rhone Valley to France and Spain	Originates in the Alps and moves over France and Spain towards the Mediterranean through the Rhone Valley, blowing from north-west to

		south-east. More common during winter. Very cold, dry, high-velocity wind - can suddenly drop temperatures below freezing. People protect orchards and gardens with thick hedges.
Bora	Dinaric Alps to the Adriatic coast, Eastern Europe	Cold winter wind (similar to the mistral) occurring in Eastern Europe. Travels from the north or north-east; can exceed 160 km/hr.

ROSSBY WAVES

Towards the mid and higher latitudes, the upper tropospheric westerlies and jet streams are wave-like, meandering winds called **Rossby waves**. The upper atmospheric winds near the polar regions, while trying to conserve angular momentum, take a wave-like, meandering path. The meandering is much more pronounced as we approach the polar regions, especially beyond **50° latitude**.

PREVIOUS YEAR QUESTIONS - PRELIMS

- Consider the following statements about the 'Roaring Forties': [2000]
 - They blow uninterrupted in the Northern and Southern Hemispheres.
 - They blow with great strength and constancy.
 - Their direction is generally from North-West to East in the Southern Hemisphere.
 - Overcast skies, rain and raw weather are generally associated with them.
 Which of these statements are correct?
 (a) I, II and III (b) II, III and IV (c) I, III and IV (d) I, II and IV
- Westerlies in the southern hemisphere are stronger and persistent than in the northern hemisphere. Why? [2011]
 - The southern hemisphere has less landmass as compared to the northern hemisphere.
 - Coriolis force is higher in the southern hemisphere as compared to the northern hemisphere.
 Which of the statements given above is/are correct?
 (a) 1 only (b) 2 only (c) Both 1 and 2 (d) Neither 1 nor 2
- Consider the following statements:[2015]
 - The winds which blow between 30°N and 60°S latitudes throughout the year are known as westerlies.
 - The moist air masses that cause winter rains in North-Western region of India are part of these westerlies.
 Which of the statements given above is/are correct?
 (a) 1 only (b) 2 only (c) Both 1 and 2 (d) Neither 1 nor 2
- With reference to "Coriolis force", which of the following statements is/are correct? [2024]
 - It increases with increase in wind velocity.
 - It is maximum at the poles and is absent at the equator.
 Select the answer using the code given below:
 (a) 1 only (b) 2 only (c) Both 1 and 2 (d) Neither 1 nor 2

PREVIOUS YEAR QUESTIONS - MAINS

- Major hot deserts in the northern hemisphere are located between 20-30 degrees north and on the western side of the continents. Why? [2013]

NCERT Reference

NCERT Class 11 Fundamentals of Physical Geography	Chapter 9: "Atmospheric Circulation and Weather Systems"
---	--